Seasonality and Immunity in Disease Dynamics

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Background

Most models of disease transmission dynamics do not consider time dependent parameters. In some systems, the effects of temperature and other time-varying variables may be too strong to ignore. In this work, we consider a model taking into account *Daphnia dentifera* recovery from *Metschnikowia bicuspidata* infections [8] and the role of temperature [5] in population dynamics.

Daphnia dentifera is a model organism that has been used for decades in epidemic research. However, there are still many open questions regarding whether an epidemic will occur or not, and if yes, what the disease prevalence will be. Due to the importance of this organism in ecology, epidemiology and evolution, sufficient data exist [7] for epidemic models to be parameterized and compared to. The available data sets suggest that there exists a large variation in observed prevalence that has not yet been explained by models.

Goal: Develop a model which allows infected hosts I to recover and move back into the susceptible class S and incorporates temperature dependence to reflect the epidemic patterns observed in field studies [7].

Disease System

Daphnia dentifera are filter-feeders and may become infected by the fungal pathogen *Metchnikowia bicuspidata* when they accidentally ingest its spores. It has recently been determined that they are able to recover [7] from the infection. In addition, it is known that several processes, such as host feeding rate, parasite production, and parasite infectivity, depend on temperature [6]. We use our model to investigate whether temperature dependence and/or recovery can explain the emergence of epidemics in late fall.

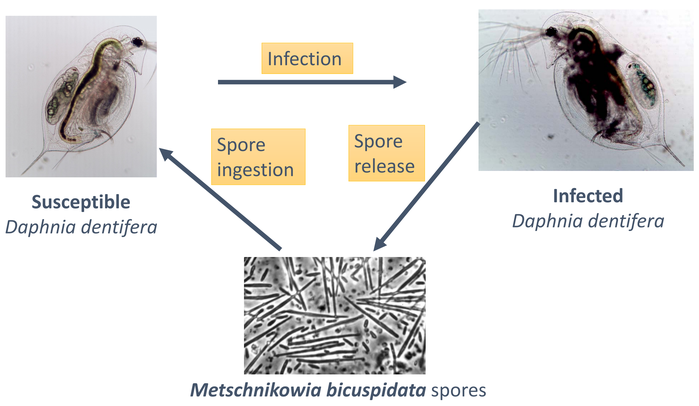


Figure 1: *Susceptible hosts may become infected after ingesting fungal spores existing in the water column. Spores are released back into the water column by dead infected Daphnia. Pictures obtained from [1], [4]*

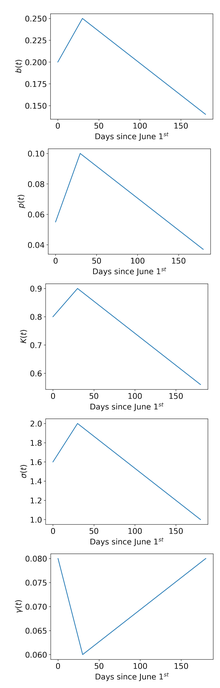
Model

$$\frac{dS}{dt} = b(t)(S+\rho I)\left(1-\frac{S+I}{K(t)}\right) - (d+p(t))\,S - \beta SZ + \gamma(t)kI$$

$$\frac{dI}{dt} = \beta SZ - [d+(1-k)v+\theta p(t)]\,I - \gamma(t)kI$$

$$\frac{dZ}{dt} = \sigma(t)\,[d+(1-k)v]\,I - \lambda Z - f\,[S+I]\,Z$$

Susceptible *Daphnia* S grow at a rate $b(t)$ and have carrying capacity $K(t)$, where t denotes time. They die at a background mortality rate d and due to predation by fish at a rate $p(t)$. They get infected at a rate β as they ingest fungal spores Z. After spore ingestion, they move into the infected *Daphnia* class, I. Infected *Daphnia* have lower fecundity, quantified by the parameter ρ, and die at an increased rate, denoted by $\theta > 1$, due to selective fish predation. A proportion $1-k$ of them dies at a rate v due to disease-induced mortality. The remaining proportion k recovers and moves back into the susceptible class S at a rate $\gamma(t)$ [8]. Fungal spores are released in the water column by dead infected hosts with a proportionality function $\sigma(t)$. They are eliminated from the water column due to sinking and UV destruction at a rate λ and are also eliminated when *Daphnia* ingest them at a constant rate f. Birth rate, predation, and the spore release parameter are positively correlated with temperature, while the recovery rate is negatively correlated with temperature.



Model variables & parameters		
Symbol	Definition	Value
b	growth rate	0.2 day^{-1} [2]
ρ	decreased fecundity parameter	0.3-0.9 [2, 3]
K	carrying capacity	0.1-0.8 mg C/l [3]
d	background mortality rate	0.03 day^{-1} [2]
p	predation rate	0.05 day^{-1} [2]
β	transmission rate	1-3 l (mg C day)$^{-1}$ [3]
k	fraction of recovering hosts	0.3 [8]
v	disease induced mortality rate	0.08 day^{-1} [2]
θ	fish predation constant	3 [2]
γ	recovery rate	0.1 days^{-1} [8]
σ	spores per dead infected host	3-5 mg C/l l/mg C [3]
λ	spore loss rate	0.05 day^{-1} [2]
f	spore removal rate by hosts	0.1-0.2 l (mg C day)$^{-1}$ [3]

Simulations

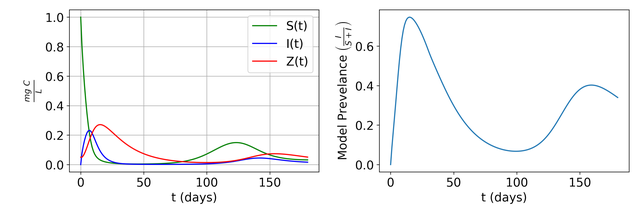


Figure 4: *Example Endemic Dynamics*

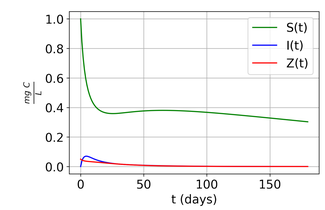


Figure 5: *Example Disease Free Dynamics*

Future Directions

- Consider other seasonality effects, if temperature alone is insufficient.
- Determine parameters such that simulated prevalence reflects observed data [7]

Acknowledgements

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References

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